np. (Packages).

Sequence Numbers → arange. / linspace } = 1D array.

matplotlib.
↓
pyplot.

np → random.

Random Numbers → random_sample(shape) - / randint(start, stop, shape) } = nD array.

import numpy as np.
np.random.random_sample.

import numpy.random as np.
np.random_sample( ) ✓

[10, 20, 30, 40, 50] = 1D ⇒ shape ⇒ (5,)

2D = Set of 1D arrays (5,) - 1D

[[10, 20, 30, 40, 50],
[60, 70, 80, 90, 100]] = (2, 5) → 2D

3D array: Set of 2D arrays

[[[9, 16, 8, 12, 8],
[1, 8, 12, 0, 16], (3,5)
[2, 5, 7, 0, 8]],

[[1, 8, 2, 6, 4],
[8, 7, 8, 1, 2], (3,5)
[0, 1, 8, 7, 6]]]

⇒ (2, 3, 5) ✓

(Deep learning)

5 Students Mark.

maths = [86, 89, 92, 90, 85]
Physics = [92, 90, 81, 86, 94]
chemistry .. [84, 92, 86, 87, 87]

marks = [[86, 89, 92, 80, 85],
[92, 90, 81, 86, 94],
[84, 92, 86, 87, 87]]

(3, 5)
↓ Subject
→ student

[[[86, - - 85],

[[[86, - - 85],
[92. - 94],
[84, - 87]],

[[86, -- 86],
[90 - — -]
[. - -]],

[[— — -]
— — — .]],

[[— — -],
[. — —]],]

Subject

(4 nay)

5D array → (5, 3, 2, 4, 6)

(4, 3, 5) = 3D array

TEST Subject Student

(300, 3)

[[[247, 248, 16], [158, 200, 240], . - - [300]]
[[- — —] [— — 7 — — [ ]]]
[[-- — —]—[- —] - - [ ])]]

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Slicing and indexing

Monday - Sunday